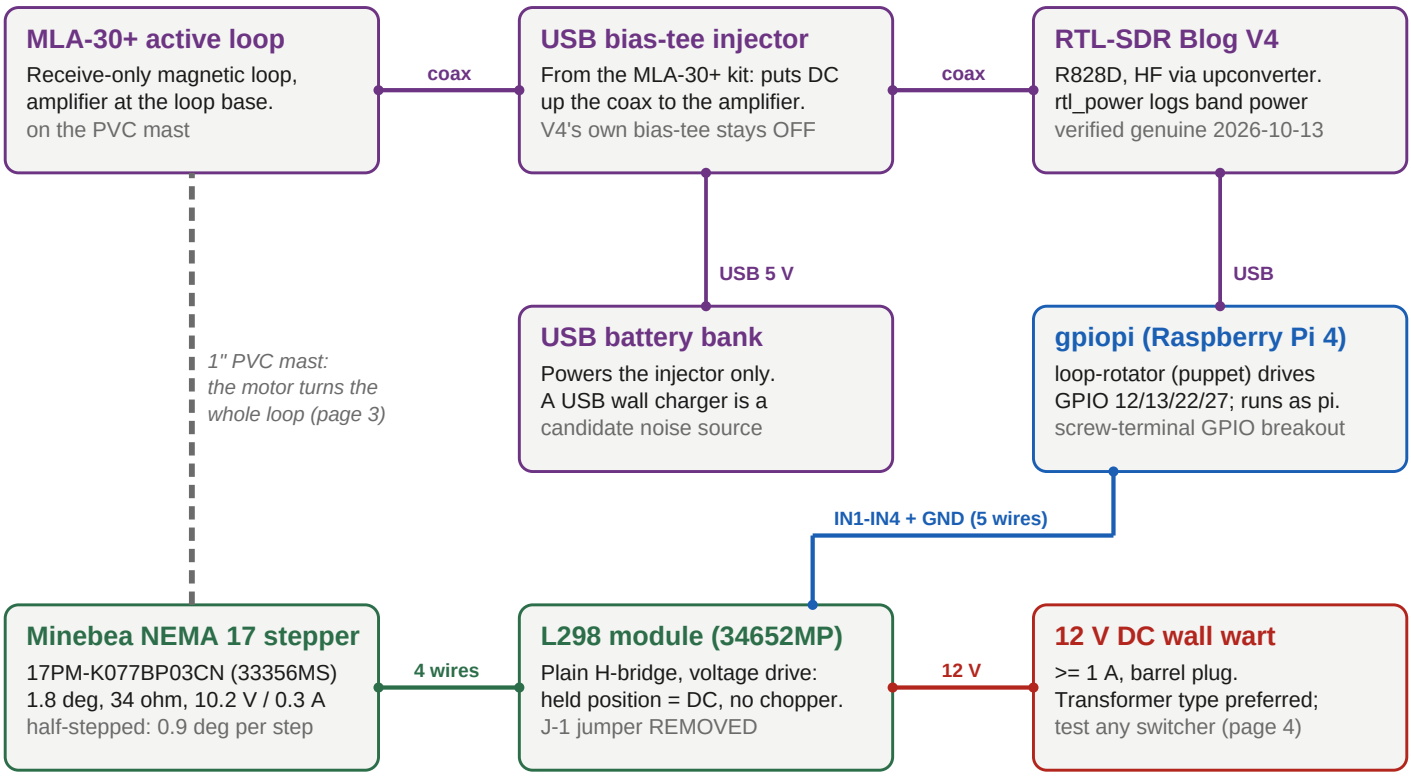
Overview and parts

A stepper turns the loop so gpiopi can take null bearings unattended (sdr-interference-hunting.md).



Parts

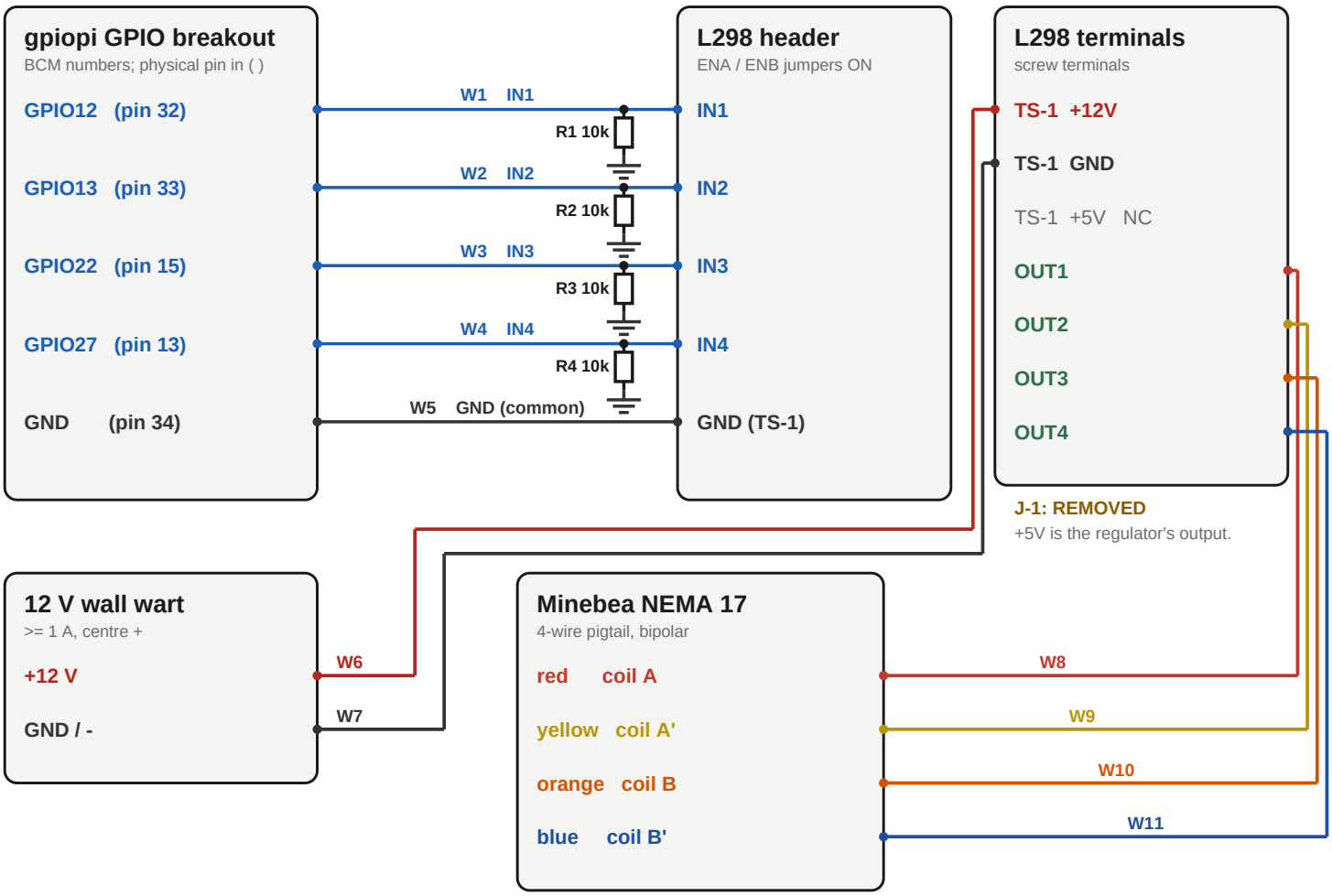
Qty	Part	Source / notes
1	Minebea 17PM-K077BP03CN NEMA 17 stepper	MPJA 33356MS (on hand)
1	L298 stepper / DC driver module	MPJA 34652MP (on hand)
4	10 kohm resistor, 1/4 W	IN1-IN4 pull-downs (page 2)
1	12 V DC >= 1 A supply, barrel plug + screw-terminal adapter	e.g. MPJA 19452PL plug-to-terminal
1	Flexible beam coupler, 5 mm to 8 mm (or 5/16")	motor shaft to mast plug
1	Hardwood dowel, 1" x 2-3", and 8 mm or 5/16" bolt / rod	plug for the bottom of the mast
1	NEMA 17 L-bracket, M3 x 6-8 mm screws	motor to base board
1	Base board (plywood) + 2 spacer blocks (2x4 offcuts)	page 3
1	Hose clamp to fit 1" PVC	thrust collar on top of the 4x4
2-3	Clip-on ferrite cores for the coax	at the amplifier and the mast foot
~6	Hook-up wire (22-24 AWG), ideally 5 colours	Pi to L298, and ground

Three things that break hardware

1. J-1 on the L298 module must be REMOVED. It ties the logic 5 V rail to the motor supply; fitted with 12 V applied, it puts 12 V into the L298's logic and out of its +5V terminal.
2. The L298 module's +5V terminal is an OUTPUT (78M05 regulator). Connect nothing to it, and never connect the Pi's 5 V anywhere on the module.
3. Do not unplug the motor while it is powered: the L298 is driving 60 mH coils.

Electrical wiring

Every line is one physical wire, numbered W1-W11 to match the schedule below.



Every ground symbol is this one common ground: Pi GND = L298 GND = supply -. R1-R4 sit at the L298 end of W1-W4.

Wire schedule

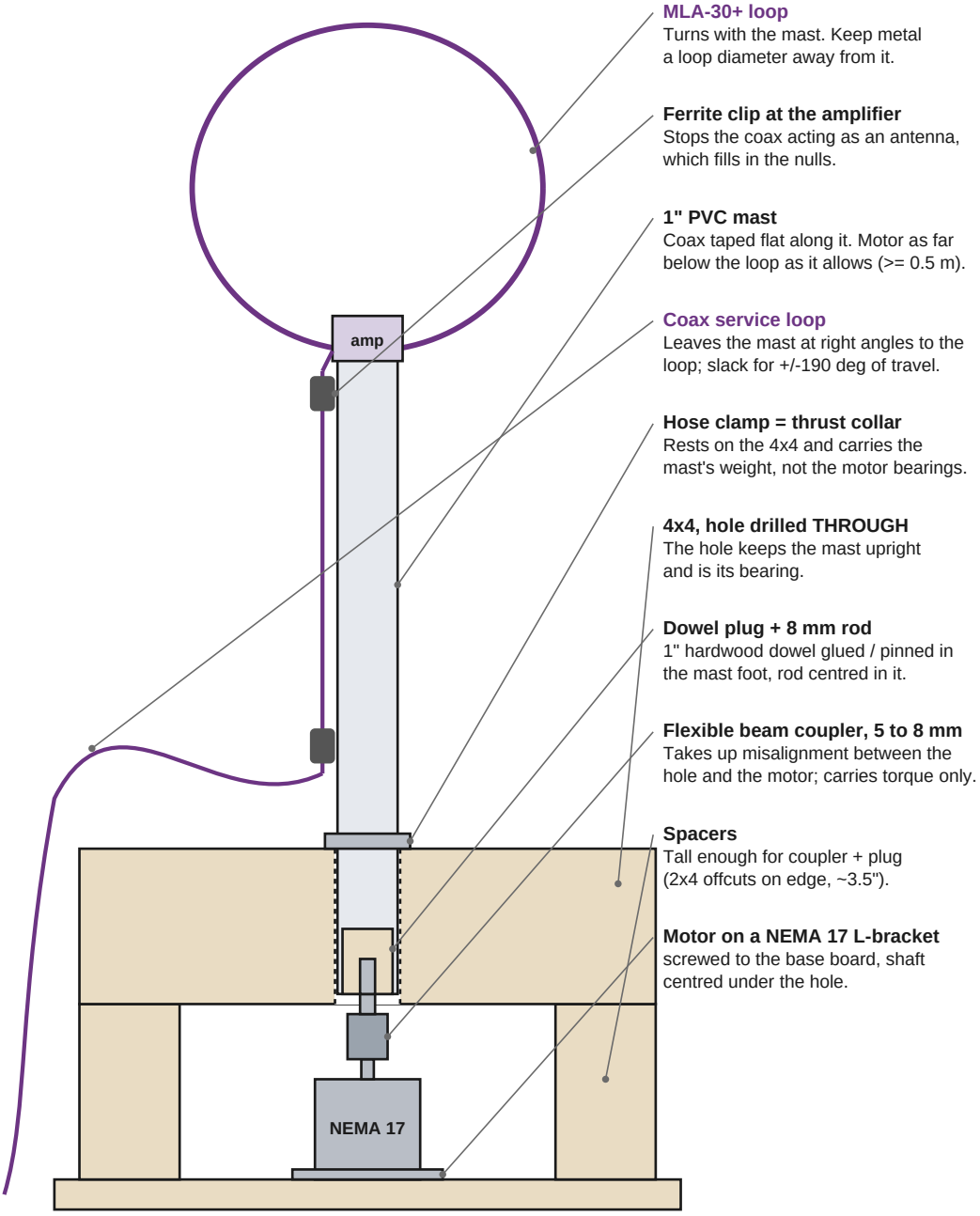
Wire	From	To	Notes
W1	gpiopi GPIO12 (pin 32)	L298 IN1	coil A side 1
W2	gpiopi GPIO13 (pin 33)	L298 IN2	coil A side 2
W3	gpiopi GPIO22 (pin 15)	L298 IN3	coil B side 1
W4	gpiopi GPIO27 (pin 13)	L298 IN4	coil B side 2
W5	gpiopi GND (pin 34; or 6/9/14/20/25/30/39)	L298 TS-1 GND	REQUIRED: logic reference
W6	wall wart +12 V	L298 TS-1 +12V	check polarity with a meter
W7	wall wart GND	L298 TS-1 GND	shares the terminal with W5
W8	L298 OUT1	motor red	OUT1/OUT2 = 'OUT A' terminal
W9	L298 OUT2	motor yellow	
W10	L298 OUT3	motor orange	OUT3/OUT4 = 'OUT B' terminal
W11	L298 OUT4	motor blue	
R1-R4	10 kohm from each of IN1-IN4	to GND	at the L298 end (note below)

Why the pull-down resistors

The L298 module has no pull resistors on IN1-IN4 (MPJA schematic), and an undriven L298 input can read high and switch a coil on. gpiopi pulls the pins down at boot (config.txt) and loop-rotator re-applies that on every exit, but R1-R4 keep the coils off even when the Pi is off, rebooting or unplugged. Fit them at the L298 end of W1-W4 where you can.

Mechanical setup

Side view, not to scale. The 4x4's hole stays the bearing; the motor only turns the mast.



MLA-30+ loop
Turns with the mast. Keep metal a loop diameter away from it.

Ferrite clip at the amplifier
Stops the coax acting as an antenna, which fills in the nulls.

1" PVC mast
Coax taped flat along it. Motor as far below the loop as it allows (>= 0.5 m).

Coax service loop
Leaves the mast at right angles to the loop; slack for +/-190 deg of travel.

Hose clamp = thrust collar
Rests on the 4x4 and carries the mast's weight, not the motor bearings.

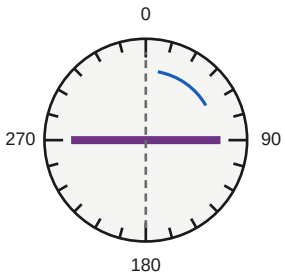
4x4, hole drilled THROUGH
The hole keeps the mast upright and is its bearing.

Dowel plug + 8 mm rod
1" hardwood dowel glued / pinned in the mast foot, rod centred in it.

Flexible beam coupler, 5 to 8 mm
Takes up misalignment between the hole and the motor; carries torque only.

Spacers
Tall enough for coupler + plug (2x4 offcuts on edge, ~3.5").

Motor on a NEMA 17 L-bracket
screwed to the base board, shaft centred under the hole.



Top view: base plate

Angles increase clockwise (blue).
Purple: the loop, edge-on, at 0 deg.
Dashed: loop axis = null line, on 0 deg.

Reading a bearing

A small loop nulls when it FACES the source: the source lies along the loop's axis (dashed), the line through the hole. Zero the rotator with that axis on 0 deg, and the null angle that loop-rotator reports IS the source's plate bearing, or the opposite one (analyze prints both). Resolve which with a second bearing from another spot: the lines cross on the source. Note the compass direction of plate 0 deg before the first sweep, to turn plate angles into real bearings.

Bring-up and first sweep

Tick each step. Commands run on gpiopi as pi (ssh gpiopi); no root needed.

A. Before any power

- [ ] Meter the motor pigtail: red-yellow and orange-blue each about 34 ohm; red-orange open. If not, the coil pairs differ from page 2: find the pairs by resistance and wire each pair to one OUT terminal.
- [ ] J-1 removed from the L298 module. ENA and ENB jumpers fitted.
- [ ] R1-R4 (10k) fitted from IN1-IN4 to GND.
- [ ] Wall wart polarity checked with a meter before it goes into TS-1.

B. Power, motor not yet connected

- [ ] Apply 12 V only. TS-1 +5V should read about 5 V; IN1-IN4 about 0 V; module LED on.
- [ ] Connect W1-W5 to gpiopi. IN1-IN4 still about 0 V.
- [ ] Power off, connect the motor (W8-W11), power on. The motor should be free to turn by hand.

C. Software

- [ ] Puppet has run on gpiopi since the loop-rotator commit:
  ```
  which loop-rotator
  ```
- [ ] By hand, aim the line through the loop's hole at 0 on the plate, then:
  ```
  loop-rotator zero
  loop-rotator goto 45
  ```
- [ ] Plate reads 45, turned clockwise? If it went to 315 instead:
- [ ] Put it back and declare the direction reversed:
  ```
  loop-rotator goto 0
  loop-rotator zero --reverse
  ```
- [ ] Back and forth a few times; it must come back to 0 exactly:
  ```
  loop-rotator goto 180
  loop-rotator goto 0
  ```

D. Is the motor kit quiet?

- [ ] Log the AM band (Ctrl-C to stop):
  ```
  rtl_power -f 520k:1720k:2k -g 20 -i 1 noise.csv
  ```
- [ ] Meanwhile, ~30 s each, noting the times: wart unplugged; wart plugged in, coils off; then coils holding, which only happens during a sweep's dwell:
  ```
  loop-rotator sweep --stop 0 --dwell 30
  ```
- [ ] The floor at a quiet channel (720 kHz) should not move between the three. If it does: ferrite on the DC lead, then another wart.

E. Calibration sweep on an AM station

- [ ] Two passes over 180 deg in 9 deg steps (about 9 minutes):
  ```
  loop-rotator sweep --stop 180 --step 9 \
    --dwell 10 --passes 2 \
    --rtl-power ~/cal.csv
  ```
- [ ] Then (the event log path is printed by the sweep):
  ```
  loop-rotator analyze \
    --log ~/loop-rotator/sweep-*.log \
    --rtl ~/cal.csv --freq 680k \
    --freq 750k --ref 720k
  ```

What good looks like

Null depth over 20 dB, and both passes agree within a few deg.
680 and 750 kHz null at similar angles if they are co-sited.
Under ~10 dB: coax pickup. Add ferrites, re-route, repeat.
Passes disagree: lost steps. Lower --rate, check the coupler.

Troubleshooting

Symptom	Likely cause	Fix
Motor buzzes or twitches, does not turn	A coil's pair split across OUT A / OUT B	Re-pair by resistance (step A1)
Turns the wrong way	One coil's two wires swapped (harmless)	zero --reverse (step C)
Misses steps / passes disagree	Too fast for the loop's inertia, or binding	--rate 10; check coupler + collar
L298 or motor hot after long holds	Expected: ~6 W while holding	Coils release after each command
'position unknown'	No zero recorded yet	Set to 0 by hand; loop-rotator zero
'another loop-rotator is running'	A sweep is still going	Wait, or stop it with Ctrl-C
rtl_power: device busy / exited	Another rtl_* program has the dongle	Stop it; one user at a time
All channels identical in analyze	rtl_power span under ~1 MHz	Keep the default 520k:1720k span